CAPES-COFECUB Program

Project: "Extreme Events and Climate Change: Impacts on the Cryosphere from the Tropical to the Polar Regions (TROPIPOLAR-CRYOEXTREMES)"

COURSE SCHEDULE

International Training Course on the Atmospheric Boundary Layer and Energy Balance over Glacier and Ocean Surfaces

19–23 October 2026
Institute of Oceanography – Federal University of Rio Grande (FURG)
Rio Grande, RS, Brazil

Languages & Communication

English, Portuguese and Spanish are all welcome!

While most lectures and practical activities will be conducted in English, participants are encouraged to ask questions, share ideas and join discussions in whichever of these three languages they feel most comfortable using.

Don't worry about speaking perfect English! What matters most is learning together, exchanging experiences and enjoying the international atmosphere of the course!

Monday, 19 October

Morning Venue: João Rocha Auditorium – Institute of Oceanography (IO/FURG)

09:30–09:40 | Official Opening and Welcome Remarks

Opening and Introduction: *Prof. Dr. Jorge Arigony-Neto*
Course Coordinator – Federal University of Rio Grande (FURG)

Institutional Welcome: *Prof. Dr. Dione Kitzmann*
Director of the Institute of Oceanography – FURG

09:40–10:10 | Invited Lecture: An Overview of Glaciological and Meteorological Studies in the Tropical Andes
Speaker: *Prof. Dr. Jean Emmanuel Sicart*
Institut des Géosciences de l'Environnement (IGE), Université Grenoble Alpes, France

An overview of glaciological and meteorological research in the tropical Andes, highlighting glacier–atmosphere interactions and observational studies.

10:10–13:00 | Introduction to the Atmospheric Boundary Layer
Lecturer: *Prof. Dr. Jean Emmanuel Sicart*

Fundamentals of the atmospheric boundary layer; surface–atmosphere interactions; energy and turbulent fluxes.

Afternoon Venue: Room S3 – Institute of Mathematics, Statistics and Physics (IMEF/FURG)

14:30–17:30 | Convective and Stable Boundary Layers
Lecturer: *Dr. Marcelo Freitas Santini, with contributions from Prof. Dr. Jean Emmanuel Sicart*

Convective and stable boundary-layer processes; turbulence and atmospheric stability.

18:00–20:00 | Icebreaker & Welcome Reception

Venue: *FURG Nautical Museum (Museu Náutico da FURG)*

Hosts: *Prof. Dr. Dione Kitzmann, Prof. Lauro Barcellos, and Prof. Dr. Jorge Arigony-Neto*

Informal gathering to welcome participants and instructors, encourage networking, and promote interaction among the international course community.

Shuttle transportation will be provided from FURG to the Nautical Museum and from the museum to Cassino.

Tuesday, 20 October

Venue: Room S3 – Institute of Mathematics, Statistics and Physics (IMEF/FURG)

09:30–12:30 | Atmospheric Boundary Layer over the Ocean
Lecturer: *Dr. Marcelo Freitas Santini*

Air–sea interactions; marine boundary-layer processes; surface energy and turbulent fluxes over the ocean.

14:30–17:30 | Atmospheric Boundary Layer over Glaciers and Snow
Lecturer: *Prof. Dr. Jean Emmanuel Sicart*

Glacier and snow surface energy balance; boundary-layer processes over ice and snow; characteristics of polar and glacier environments.

Wednesday, 21 October

Venue: Room S3 – Institute of Mathematics, Statistics and Physics (IMEF/FURG)

09:30–12:30 | Boundary Layers in Different Environments
Lecturers: *Prof. Dr. Jean Emmanuel Sicart and Dr. Marcelo Freitas Santini*

Comparison of ocean, glacier, snow and other surface environments; similarities and contrasts in surface–atmosphere exchanges.

14:30–17:30 | Measurement Principles and Instrumentation
Lecturer: *Dr. Marcelo Freitas Santini*

Micrometeorological measurements; meteorological and radiation sensors; turbulent flux measurements; field instrumentation and observational strategies.

17:30–17:50 | Snack Break – FURG Student Center (Centro de Convivência)

Short break for participants to purchase snacks and refreshments at their own expense.

17:50–18:15 | Transfer to the FURG Oceanographic Museum

18:15–20:00 | Night at the Museum – Visit to the FURG Oceanographic Museum
Venue: *FURG Oceanographic Museum (Museu Oceanográfico da FURG)*

Guided visit to the Oceanographic Museum of FURG, providing an opportunity to explore its collections and exhibitions, learn about marine science and oceanographic research, and promote informal interaction among course participants and instructors.

Shuttle transportation will be provided from the FURG Student Center to the Oceanographic Museum and from the museum to Cassino.

Thursday, 22 October

Venue: Room S3 – Institute of Mathematics, Statistics and Physics (IMEF/FURG)

09:30–12:30 | Data Analysis and Case Studies I
Instructors: *Dr. Christian Torres, Prof. Dr. Jean Emmanuel Sicart and Dr. Celina Cândida Ferreira Rodrigues*

Introduction to observational datasets; data processing and quality control; analysis of atmospheric and surface-energy variables.

14:30–17:30 | Data Analysis and Case Studies II
Instructors: *Dr. Christian Torres, Prof. Dr. Jean Emmanuel Sicart and Dr. Celina Cândida Ferreira Rodrigues*

Hands-on analysis of observational datasets; surface energy balance and turbulent flux calculations; interpretation and discussion of results.

Friday, 23 October

Morning Venue: Room S3 – Institute of Mathematics, Statistics and Physics (IMEF/FURG)

09:30–11:30 | Data Analysis and Case Studies III
Instructors: *Dr. Christian Torres, Prof. Dr. Jean Emmanuel Sicart and Dr. Celina Cândida Ferreira Rodrigues*

Integrated analysis of ocean and glacier datasets; comparison of different environmental conditions; synthesis and discussion of case studies.

11:30–12:30 | Lunch Break

12:30–13:00 | Transfer from FURG to CCMAR/FURG

13:00–18:00 | Field Activity and Course Wrap-up
Meeting and departure point: *CCMAR/FURG – Centro de Convívio dos Meninos do Mar*

Lead Faculty: *Prof. Dr. Osmar Möller*

Field excursion in the Patos Lagoon estuary, including observations of the estuarine environment, discussion of atmosphere–ocean interactions, and practical aspects of field measurements and observational procedures.

Final discussion and course wrap-up.

18:00–18:30 | Disembarkation and Preparation for Departure

18:30 | Shuttle Departure – CCMAR/FURG to Cassino

Transportation Schedule – Shuttle Service

Shuttle transportation will be provided for course participants between Cassino, the FURG campus, and the venues of the social, cultural, and field activities, according to the schedule below.

Date	***Departure***	***Route***
Monday, 19 October	*08:45*	*Cassino → FURG*
	17:30	*FURG → FURG Nautical Museum*
	20:00	*FURG Nautical Museum → Cassino*
Tuesday, 20 October	*08:45*	*Cassino → FURG*

	17:30	*FURG → Cassino*
Wednesday, 21 October	*08:45*	*Cassino → FURG*
	17:50	*FURG Student Center → FURG Oceanographic Museum*
	20:00	*FURG Oceanographic Museum → Cassino*
Thursday, 22 October	*08:45*	*Cassino → FURG*
	17:30	*FURG → Cassino*
Friday, 23 October	*08:45*	*Cassino → FURG*
	12:30	*FURG → CCMAR/FURG*
	18:30	***CCMAR/FURG → Cassino***

Departure times refer to scheduled shuttle departures. Exact boarding locations in Cassino and at FURG will be communicated to participants.

Participants are kindly requested to arrive at the designated boarding points a few minutes before departure.

Workload and Additional Activities

Academic programme: *31 hours of scheduled activities, including the official opening and invited lecture.*

Additional social and cultural activities: *3 hours and 45 minutes.*

- ***Monday, 19 October, 18:00–20:00:*** *Icebreaker & Welcome Reception – FURG Nautical Museum.*
- ***Wednesday, 21 October, 18:15–20:00:*** *Night at the Museum – FURG Oceanographic Museum.*

Main teaching venue: *Room S3, Institute of Mathematics, Statistics and Physics (IMEF/FURG).*

Opening session venue: *João Rocha Auditorium, Institute of Oceanography (IO/FURG), Monday morning.*

Field activity departure point: *CCMAR/FURG, Friday at 13:00.*

Course Organizers

- ***Jorge Arigony-Neto*** *— Federal University of Rio Grande (FURG), Brazil*
- ***Luciano Pezzi*** *— National Institute for Space Research (INPE), Brazil*
- ***Marcelo Freitas Santini*** *— National Institute for Space Research (INPE), Brazil*
- ***Celina Cândida Ferreira Rodrigues*** *— National Institute for Space Research (INPE) and Rio de Janeiro State University (UERJ), Brazil*
- ***Christian Torres*** *— National Institute for Space Research (INPE), Brazil*

- ***Jean Emmanuel Sicart*** *— Institut des Géosciences de l'Environnement (IGE), Université Grenoble Alpes (UGA) / Institut de Recherche pour le Développement (IRD), France*

This is a preliminary schedule and may be subject to minor adjustments by the Course Organizing Committee.